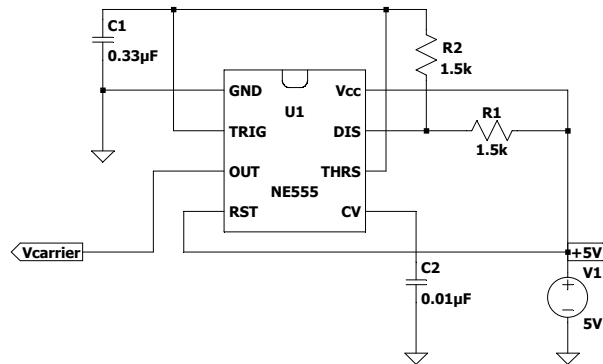
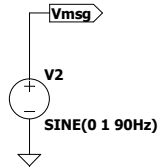


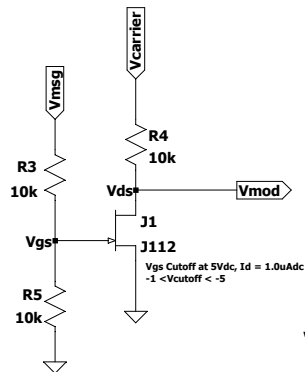
MESSANGER

Carrier Signal

Message Signal



Amplitude Modulation



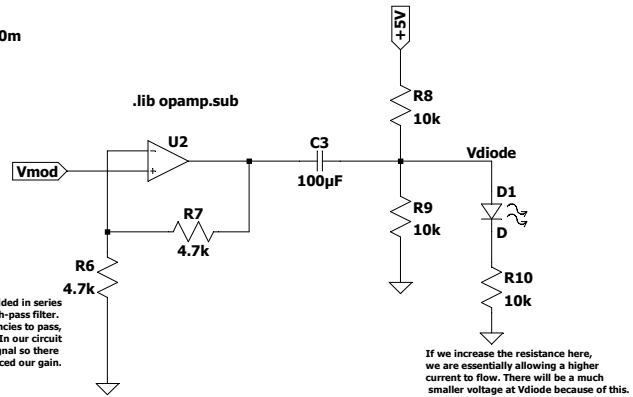
N Channel
- electrons are charge carriers
- gate has a p type
- drain and source have an n type

Depletion mode
- gate is normally closed, allowing current to flow. This is done via the drain and source sharing the same n type material.

Depletion region increases as Vgs becomes more negative.

FINAL STAGE(FOR SENDING SIGNAL)

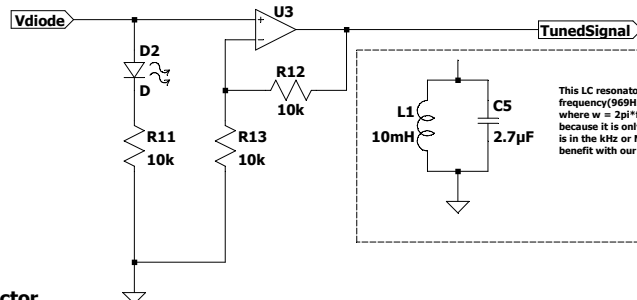
.tran 100m



If a capacitor is added in series with R6 to GND, it'll form a high-pass filter. Allowing only high frequencies to pass, blocking low frequencies. In our circuit we have a low frequency signal so there was some impedance, which reduced our gain.

RECEIVER

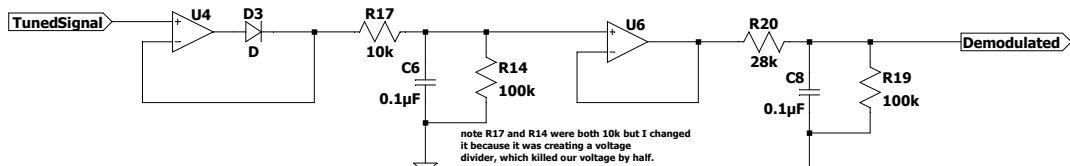
Pick Frequency Bandwidth



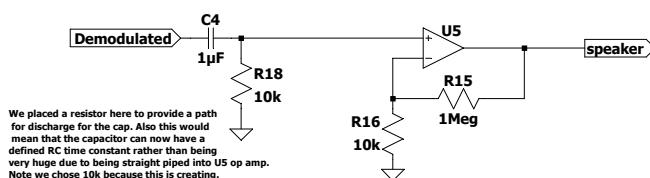
Peak Detector

aka envelope detection via rectification(uses diode)

need this unity gain voltage follower because if not, the voltage from the first stage can be drooped down by the second capacitor drawing current to charge itself.



AC coupling(DC removal)



note that R17 and R20 serve the purpose to get rid of our ideal op amps case. Since the op amps we are using in LTSPICE are ideal, they have a property called infinite current sourcing capability. This means that it can supply infinite current at the output without any voltage drop due to zero output impedance.